SAHAYANATHAN RENOLSON

ELECTRICAL & ELECTRONIC ENGINEERING

(UNDERGRADUATE)

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LinkedIn | Portfolio | GitHub

PROFESSIONAL SUMMARY

Electrical and Telecommunication Engineering undergraduate with hands-on experience in industrial automation, embedded systems, IoT, and robotics. Experienced with Siemens PLC/HMI platforms, ESP32-based control, cloud-connected sensing, serial communication, and robot kinematics. Industrial training at Nikini Automation Systems and SLT Digital Lab strengthened practical skills in commissioning, troubleshooting, R&D, and end-to-end engineering integration. Currently developing a 12-DOF quadrupedal robot as a final-year project with embedded control, IMU sensing, MQTT telemetry, MATLAB modelling, and ROS2-ready expansion.

EDUCATION

B.Sc. Eng. (Hons) in Electrical and Electronic Engineering — **2022 - 2026 (Expected)**

South Eastern University of Sri Lanka | Oluvil, Ampara, Sri Lanka

Focus: Embedded Systems, Industrial Automation, Robotics, IoT, Control, and Telecommunication Engineering

PROFESSIONAL EXPERIENCE

Automation Engineer Trainee — **Aug 2025 - Nov 2025**

Nikini Automation Systems (Pvt) Ltd. - Control / Electrical Division Bandaragama, Sri Lanka

• Developed and debugged PLC/HMI logic using Siemens S7-300 and S7-1200 platforms, Siemens KTP Basic panels, and TIA Portal V13-V20.

• Configured RS-232 and RS-485 (Modbus) communication and integrated load-cell weighing indicators, industrial sensors, and data-logging workflows.

• Contributed to a complete weighing automation project covering PLC sequencing, HMI real-time monitoring, RS-485 integration, and data transfer to a database/web portal.

• Assisted with industrial vision-camera installation and commissioning, PLC hardware interfacing, generator-controller integration, preventive maintenance, diagnostics, and PID-control activities.

• Gained basic practical exposure to fiber-optic splicing using the Comway C10 Arc Fusion Splicer.

Electrical & Electronic Engineer Trainee — **Jun 2024 - Sep 2024**

SLT Digital Lab - Sri Lanka Telecom, Research & Development Section

Colombo, Sri Lanka

• Gained practical experience in telecommunications R&D, IoT systems, sensor integration, cloud-connected monitoring, and network-infrastructure development.

• Contributed to the Smart Agro project using AWS IoT Core, MQTT, Grafana, Python, EC2, and S3 for real-time field monitoring and smart-irrigation workflows.

FEATURED FINAL-YEAR PROJECT

Development of a 12-DOF Quadrupedal Robot with Enhanced Navigation | *3-member team | Ongoing*

• Designed and assembled a locally fabricated 12-DOF quadruped with four 3-DOF legs (hip roll, hip pitch, knee pitch), 12 Feetech STS3215 serial-bus servos, and FE-URT-1 communication.

• Developed forward/inverse kinematic models, linkage-based knee analysis, CAD/3D-printed structure, and MATLAB leg-trajectory/gait simulations before hardware testing.

• Implemented ESP32-based real-time motion control, remote mode selection, servo feedback monitoring, MPU-6050 6-axis IMU sensing, and MQTT telemetry/dashboard integration.

• Validated standing, sitting, forward/backward walking, body sway, and front/rear push-up motions; developed initial URDF/Gazebo/ROS2 integration for future higher-level control and navigation research.

SELECTED PROJECTS

Industrial Weighing Automation System | *Nikini Automation Systems*

• Contributed to PLC program development and commissioning for machine sequencing, logic control, and data logging using Siemens automation platforms.

• Created/deployed HMI functions for machine control and real-time monitoring, and configured RS-485 (Modbus) communication with weighing indicators and database/web-portal integration.

Smart Agro - IoT Smart Irrigation System | *SLT Digital Lab*

• Designed and prototyped a field-monitoring and irrigation workflow using sensors, MQTT, AWS IoT Core, EC2/S3, Grafana, and Python for remote visualization and control.

Pill Pal - Automated Pill Dispenser | *Academic / Personal Project*

• Developed an IoT-based medication dispenser using NodeMCU/Arduino, sensors, actuators, C++, alert notifications, and 3D-printed hardware for scheduled pill dispensing.

TECHNICAL SKILLS

Industrial Automation & Control: Siemens TIA Portal V13-V20; S7-300; S7-1200; Siemens KTP Basic HMI; PLC/HMI logic; ladder logic; PID control; commissioning; diagnostics; preventive maintenance; load-cell weighing systems; industrial vision systems.

Industrial Communication: RS-232; RS-485; Modbus; serial communication; PLC hardware interfacing; weighing-indicator integration; database/web-portal data integration.

Embedded Systems & Robotics: ESP32; Arduino; NodeMCU; Raspberry Pi; NVIDIA Jetson Nano; STS3215 serial-bus servos; FE-URT-1; MPU-6050; sensors and actuators; Embedded C/C++; Python; MicroPython.

Robotics & Simulation: ROS / ROS2; MATLAB; forward and inverse kinematics; gait development; URDF; Gazebo; motion control; servo feedback; IMU integration.

IoT, Cloud & Data: MQTT; AWS IoT Core; AWS EC2; AWS S3; Grafana; DynamoDB; MongoDB; MySQL; PostgreSQL; PgAdmin; real-time dashboards and data logging.

Engineering Design & Development: SolidWorks; AutoCAD; EasyEDA; schematic/PCB design; 3D printing; Git/GitHub; VS Code; Arduino IDE; Linux/Ubuntu.

LANGUAGES

Tamil - Native / Fluent | **English** - Professional working proficiency | **Sinhala** - Basic conversational proficiency

REFEREES

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